Differentiation

Section 1: Product & Quotient Rules

1 Differentiate $y = x^3 e^{2x}$ with respect to x.

a. $x^2 e^{2x}(3 + 2x)$

b. $3x^2 e^{2x}$

c. $2x^3 e^{2x}$

d. $x^2 e^{2x}(3x + 2)$

2 Differentiate $y = \dfrac{\ln x}{x^2}$ for $x > 0$.

a. $\dfrac{1 - 2\ln x}{x^3}$

b. $\dfrac{1 + 2\ln x}{x^3}$

c. $\dfrac{1 - 2\ln x}{x^4}$

d. $\dfrac{1}{x^3}$

3 Find the gradient of the curve $y = x\sin(2x)$ at $x = \pi/4$.

a. 1

b. 0

c. $\pi/2$

d. 2

Section 2: Derivatives of Exponential, Logarithmic & Trig Functions

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4 Differentiate $f(x) = \ln(\sec x + \tan x)$.

a. $\sec x$

b. $\tan x$

c. $\sec^2 x$

d. $\sec x \tan x$

5 Differentiate $y = e^{\sin(3x)}$ with respect to x.

a. $3\cos(3x)e^{\sin(3x)}$

b. $\cos(3x)e^{\sin(3x)}$

c. $3e^{\sin(3x)}$

d. $-3\cos(3x)e^{\sin(3x)}$

6 Find $\frac{d}{dx}[\tan^{-1}(2x)]$.

a. $\frac{2}{1+4x^2}$

b. $\frac{1}{1+4x^2}$

c. $\frac{2}{1+2x^2}$

d. $\frac{2}{\sqrt{1-4x^2}}$

Section 3: Implicit Differentiation

7 Find $\frac{dy}{dx}$ for the curve $x^2 + xy + y^2 = 7$.

a. $-\frac{2x+y}{x+2y}$

b. $\frac{2x+y}{x+2y}$

c. $-\frac{2x}{2y}$

d. $-\frac{x+2y}{2x+y}$

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8 Find the gradient of the curve $x^3 + y^3 = 9xy$ at the point $(2, 4)$.

a. $\frac{4}{5}$

b. $-\frac{4}{5}$

c. $\frac{5}{4}$

d. 1

9 Find the equation of the normal to the curve $x^2 - 2xy + 3y^2 = 9$ at $(3, 2)$.

a. $y = 2$

b. $x = 3$

c. $y = x - 1$

d. $2x + y = 8$

Section 4: Parametric Differentiation

10 A curve is defined parametrically by $x = 2t^2$, $y = 4t$. Find $\frac{dy}{dx}$ in terms of t.

a. $\frac{1}{t}$

b. t

c. $\frac{2}{t}$

d. $2t$

11 A curve has parametric equations $x = a\cos\theta$, $y = b\sin\theta$. Find the equation of the tangent at parameter θ.

a. $\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$

b. $\frac{x}{a}\sin\theta + \frac{y}{b}\cos\theta = 1$

c. $\frac{x}{a}\cos\theta - \frac{y}{b}\sin\theta = 1$

d. $bx\cos\theta + ay\sin\theta = ab$

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12 Find $\frac{d^2y}{dx^2}$ for the parametric curve $x = t^2, y = t^3$.

a. $\frac{3}{4t}$

b. $\frac{3}{2}$

c. $\frac{3}{4t^2}$

d. $\frac{6}{t}$

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